

Lil' Willy 001 Accident Report

E-Town Rocket Bureau

Adopted: April 3, 2026

Flight Number: Flight #1
Crash Date: March 29, 2026
Payload: 1 egg
Impulse: 52.9 Ns
Mass: 550g

Executive Summary

The first practice flight of Lil' Willy 001 resulted in rapid unscheduled disassembly from a catastrophic failure localized in the coupler system, due to excessive friction between the coupler and upper section. Other systems also performed sub-optimally, leading to near-total destruction of the vehicle. This report analyzes the damage and recommends actions to take regarding redesign and safety process implementation.

Team Members Present During Accident

Pradyumna Manur

Class of 2026

Peter Elbakian

Class of 2027

Eli Drake

Class of 2029

Flight Events and Performance

Flight #1 was a practice flight aimed at validating vehicle stability, payload safety, flight mechanics, the avionics system, and the airbrake system. It was carrying a payload of one egg. Initial mass was approximately 550 grams, and average wind speed was about 4.8 m/s (10mph).

The launch and ascent phase was nominal, although with an unexpected small pitching maneuver after the vehicle cleared the launch rail. The airbrakes did not appear to deploy while in the visible range. At approximately $t=7s$, the ejection charge fired (~1.5s early). The rocket did not decouple and deploy the parachute, as was intended. It stayed in one piece and followed a ballistic trajectory into the ground.

Flight Damage Assessment

Findings

The rocket sustained catastrophic damage to nearly all systems and structures. As shown in the images below, nearly all systems were destroyed and deemed unfit for repair.



Lower Section with shattered Coupler and damaged fuselage



Upper Section with ripped fuselage, broken egg, destroyed avionics bay, and airbrake remnants



Undeployed parachute and airbrake piece



Another angle of the Lower Section



Scattered fuselage and airbrake pieces

Although the Aft System and its fins survived, the engine sustainer was bent into multiple angles, the top part of the lower fuselage collapsed, and the coupler fragmented into multiple pieces with fault lines running through them. The heat-resistant lining was not visible, although the gas expansion nozzle was intact. The upper fuselage was completely destroyed, the remnants of the airbrake were ejected from the rocket and had melted parts. The egg was found disintegrated in its capsule, and a carbon fiber rod was found to have penetrated deep into the capsule. Additionally, the capsule also bore burn marks. The avionics bay was fragmented and

had melt markings, and much of the wiring had disconnected. The 600maH 25C lithium-polymer battery was swollen, the altimeter broken into three pieces, and the nose cone had a carbon fiber rod lodged inside. The SD card failed to read when inserted into multiple computers.

Analysis

Melted plastic on multiple plastic parts—primarily the airbrake and avionics bay—indicate burn damage, likely from the hot 1530°C (2780°F) ejection charge gases. Given that the rocket didn't decouple on charge deployment, it is likely that the gases pushed into the upper section of the rocket, melting or blasting through the parachute retention system, airbrake, around the egg retainer, and into the avionics (though the egg retainer likely stopped most gasses because of its bulkhead-like nature). Excessive friction between the coupler and upper section likely caused the de-coupling issue, as a large amount of tape was installed.

Fractured, twisted, and ripped parts—like coupler, sustainer, and upper fuselage, respectively—indicate heavy damage sustained during impact. Using OpenRocket simulations, an un-airbraked flight would have ascended to over 274m (900ft). Assuming a descent from that height, the rocket would have impacted the ground at around 67 m/s (150mph / Mach 0.2), far exceeding design limits.

Additionally, the SD card read failure suggests a potential short-circuit or severe damage to the SD's metal pads.

The integrity of the aft system and its fins shows that this system is physically robust, while the undamaged expansion nozzle shows that the heat-resistant lining works.

Probable Cause

Excessive friction between the coupler and the upper stage caused by adhesive tape likely prevented separation upon deployment of the ejection charge.

Recommendations

ERB recommends the following changes to Lil' Willy and ERB's ARC 2026 program:

Critical Design Changes

- I. Redesign of the coupler and parachute deployment system:
 - A. Removal of weight-reducing gaps to increase parachute deployment reliability (reduces risk of the parachute getting caught)
 - B. Parachute stored in the bottom section (as opposed to the current top-stored one which can get stuck)
 - C. Parachute grounded to the tether between the upper and lower sections
 1. Removal of PLA current anti-twist bearing plate and implementation of smaller anti-twist bearing part at the point of parachute grounding
 - D. Simplification of coupler and, where possible, removal of 3D printed parts
 - E. Better adhesion of heat-resistant cloth through initial design considerations, rather than post-design addition
- II. Implementation of protective, sealing bulkheads:
 - A. Where gaseous forces are expected
 - B. 2-thick wooden bulkheads behind nose cone to absorb impact during high-velocity impacts (sourced from successes of ERB's ARC 2025 program)
 - C. Surrounding egg module to prevent penetration
- III. Redesign of the avionics bay to be more structurally robust
 - A. Inclusion of protective foams (with minimal effect to IMU readings)
 - B. Creation as an independent module that's safe on its own, rather than a simple part of the rocket

Recommended Design Changes

- IV. Implementation of hardware-in-the-loop software to test control system on land
- V. PCB- or perf-based board to simplify avionics assembly
- VI. Switch to acceleration-based controller start instead of time-based
- VII. Replace MicroSD card storage with 2MB or 8MB flash chips, allowing for a more robust, soldered connection and decreasing the risk of data-loss

Processes

- VIII. Implementation of ground-handling and launch procedures:
 - A. Including setup, avionics initialization, pre-flight testing, launch readiness confirmation, post-flight data recovery and handling, maintenance, and turn-around time logging
 - B. Assigning of launch checklist manager